

A Simplified 2D-3D CNN Architecture for Hyperspectral Image Classification Based on Spatial-Spectral Fusion

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Abstract—Convolutional neural networks (CNN) have led to a successful breakthrough for hyperspectral image classification (HSIC). Due to the intrinsic spatial-spectral specificities of a hyperspectral cube, feature extraction with 3D convolution operation is a straightforward way for HSIC. However, the overwhelming features obtained from the original 3D CNN network suffers from the overfitting and more training cost problem. To address this issue, in this paper, a novel HSIC framework based on a simplified 2D-3D CNN is implemented by the cooperation between a 2D CNN and a 3D convolution layer. Firstly, the 2D convolution block aims to extract the spatial features abundantly involved spectral information as a training channel. Then, the 3D CNN approach primarily concentrates on exploiting band co-relation data by using a reduced kernel. The proposed architecture achieves the spatial and spectral features simultaneously based on a joint 2D-3D pattern to achieve superior fused feature for the subsequent classification. Furthermore, a deconvolution layer intends to enhance the robustness of the deep features is utilized in the proposed CNN network. The results and analysis of extensive real HSIC experiments demonstrate that the proposed

light-weighted 2D-3D CNN network can effectively extract refined features and improve the classification accuracy.

Index Terms—CNN, Hyperspectral image classification (HSIC), Convolution, Feature extraction.

I. INTRODUCTION

Hyperspectral remote sense imaging continuously furnishes a variety of information such as radiation, space, and spectrum of features, which has an insurmountable advantage in the domain of feature recognition and classification [1-6]. HSIC aims to divide the multiple dimensional feature space into different regions with the same terrain, and has become one of the most rapid developments in earth science and remote sensing field. In the last two decades, CNNs automatically extract deep features with a hierarchical architecture have been proven to be very successful on a series of visual application and tasks, such as image denoising [7-9], image detection [10-12] and classification [13-14]. Nowadays, the existing classification methods based on the CNN framework provide rich solutions for HSIC tasks [15-36]. In general, there are three categories of the convolution operation in the existed CNN HSIC frameworks including 1D CNN, 2D CNN, and 3D CNN respectively.

The network architecture of 1D CNN is designed to use the pixel vector along the radiometric dimension as a training sample to extract deep feature [17-19], which is called spectral-based classification approach conceptually. In [17], it was the first time to employ CNN with multiple layers for HSIC directly in the spectral domain. A novel RNN model [18] was proposed to effectively analyze hyperspectral pixels as sequential data to capture the intrinsic feature, which designed a new activation function to train the network without the risk of divergence.

The 2D CNN model for HSIC is called spatial-based classification approaches tried to learn spatial features [20-24] by utilizing the similar approaches for traditional images with the colorful pattern of RGB, which brought out an inevitable drawback caused by the ignore the united spectral-spatial attributes of the specific hypercube. In [21], Hao and Wang

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designed the SRCL model for HSIC, which explored a super-resolution-aided way to construct a spatially enhanced image. In [22], a CNN-MRF model was proposed to integrate spectral and spatial information in a unified Bayesian framework by learning the posterior class distributions. Li and Xie [23] introduced a CNN model to reconstruct an enhanced image cube by bands selection with a new spatial feature-based strategy.

Since the hyperspectral image is originally 3D hypercube with the spectral and spatial continuity, the HSIC methods integrated both spectral and spatial information have gained more popularity [25-29]. Handling the hyperspectral CNN classification with 3D convolutions is a straightforward way, which is also called the spectral-spatial classification approach. In this way, 3D regions with joint spatial-spectral information can be processed simultaneously. An automatic design of CNN for the HSIC framework is explored in [27], which designed 1-D Auto-CNN and 3-D Auto-CNN to automatically extract spectral and spatial information from the original cube. Feng and Yu proposed a multiclass spatial-spectral GAN (MSGAN) method to utilize generators for the samples production and the discriminator for the joint spatial-spectral feature extraction. In [29], a semi-supervised 3D convolutional neural network for the spectral-spatial HSIC is proposed by engaging adaptive dimensionality reduction (ADR) to deal with the problem of the curse of dimensionality. Recently, a series of popular deep learning-based methods have been exploited for spatial-spectral classification. In [30], Generative adversarial networks (GAN) was employed for discriminative features extraction, 1D-GAN as a spectral classifier and a robust 3D-GAN as a spectral-spatial classifier were proposed for HSI classification respectively. In [31], a cascaded RNN model was designed to explore the redundant and complementary information of HSIs by utilizing two RNN layers. In [32], the multi-scale hierarchical recurrent neural networks (MHRNNs) was proven to be efficient in hyperspectral image classification, which learns the multi-scale local feature by 3D CNNs and learns the spatial dependency of non-adjacent image patches in the spatial domain by RNN. CSA-MSO3DCNN [33] was developed to optimize the discriminative features with attention modules and reduce the redundancy with a three-layer's octave 3D convolution. However, there are two main limitations revealed with the 3D convolution model. On one hand, with the increasing number of the 3D kernels, the complexity and time cost get higher, on the other hand, the overwhelming deep features bring out the over-fitting problem and the classification accuracy is not high actual as expected.

To alleviate the mentioned issue, in this paper, a new deep 2D-3D CNN network with the spectral-spatial fusion is explored by simplifying the original 3D network structure,

which extracted the fusion feature by jointing the spatial and spectral information simultaneously. The proposed deep network is mainly composed of two convolutional blocks for feature extraction. The first block is denoted as 2D CNN which aims to spatial information extraction with the spectral channels involved. The second block is called the 3D CNN model which focuses on integrating band information through spectral convolution to generate spectral-spatial representations of the HSI cubes with a reduced size of $1 \times 1 \times L$ filter kernel. In this framework, the proposed model extracted both spectral features and spatial neighborhood information simultaneously to improve the distinguishing ability of the fusion feature. Besides, we also adopted a deconvolution layer to solve the drawbacks of the different feature sizes and limited representation.

According to the investigation, this paper introduced the new 2D-3D structure for the HSIC for the first time, which adopted the efficient model to realize the classification with spectral-space feature fusion. The specific contributions are listed as follows:

(1) The proposed framework performs joint learning of 3D spatial-spectral representations with the flexibility of two types of kernels to achieve refined characteristics for classification. Specifically, the 3D convolution with the kernel size of $m \times m \times L$ is utilized to mine hidden feature effectively in the situation for the improvement of the classification accuracy, whereas, the simplified kernel size of $1 \times 1 \times L$ is adopted to reduce the spatial redundancy with less parameters. The relationship between the two models are the first time to be clarified that the classification performance is related to both the size of the kernel and the experimental parameters, the standard kernel performs well with more parameters and the simplified kernel is advantage for the efficiency.

(2) To our knowledge, the network is enabled to supply an implementation with the depth-wise separable convolution way for HSIC equipped with the simplified 3D kernel. Specifically, a 3D convolution with the kernel size of $1 \times 1 \times L$ is explored for the first time to implement the fusion extraction of spectral information between bands. In this way, the spectral information is extracted by the merge of the neighbor band data, and the spatial and spectral feature is fused simultaneously at the same time.

(3) To improve the efficiency of the proposed model, we employed only two convolution blocks instead of a very deep network structure. The spatial information convolution is implemented in the 2D convolution layer with a bunch of filter kernels to expand the spatial features, and the 3D block relies on only one 3D convolution kernel to increase the convolution speed and overcome the overfitting problems.

(4) Especially, we reveal the intrinsic relationship between the proposed model with simplified 3D kernel and the 1D CNN

on HSIC, and the comparisons between our model and 1D-2D CNN network are explained in this paper, which will offer a systematic reference for the learning of the 1D, 2D, and 3D CNN HSIC. Besides, we conduct experiments to exhaustively exploit the performance of different structures based on the proposed network, which will inspire the design of other CNN framework for hyperspectral applications.

The remaining part of the paper is organized as follows. The detailed of the proposed 2D-3D CNN classification network is given in Section II. Experimental results and analysis are illustrated in Section III and conclusions are drawn in Section IV.

II. CLASSIFICATION METHOD BASED ON 2D-3D CNN NETWORK

A. Network Structure of the proposed framework

The main task of hyperspectral image processing of CNN architecture is handling spatial and spectral information simultaneously via adjacent layers. In this paper, we proposed the spatial-spectral joint CNN network which contained seven layers, Fig.1 showed the hierarchical structure diagram of the proposed 2D-3D-D CNN classification framework with the specific parameters. Specifically, the proposed CNN

architecture consists of two convolution blocks (2D-CNN, 3D-CNN), the rich spatial convolutional features are obtained by 2D-CNN model, the 3D-CNN model accomplished the spectral feature processing by fused with the neighbor band information, an activator is implemented by the pooling layer in each convolution part which aims to reduce data variance and improve the nonlinearity of the feature. To maintain a good performance of the fusion of the extracted feature, a deconvolution layer is followed in the 3D-CNN part to reconstruct the feature map. Two full connection layers (FC1, FC2) are used separately to reassemble the obtained local features for the output layer. The proposed 2D-3D CNN method has the two following distinctive advantages. First, the 2D-CNN extracted rich spatial feature maps with the channel information involved. Secondly, we utilized a 3D-CNN to mine the integrated spectral-spatial characteristic. Specially, we design a simplified one filter with the size of $1 \times 1 \times L$ to exploit the spectral information in the 3D-CNN network effectively. The proposed network integrated the spatial and spectral information simultaneously with the strategy of 3D convolution. The more details of the CNN framework are introduced in the following sub-sections.

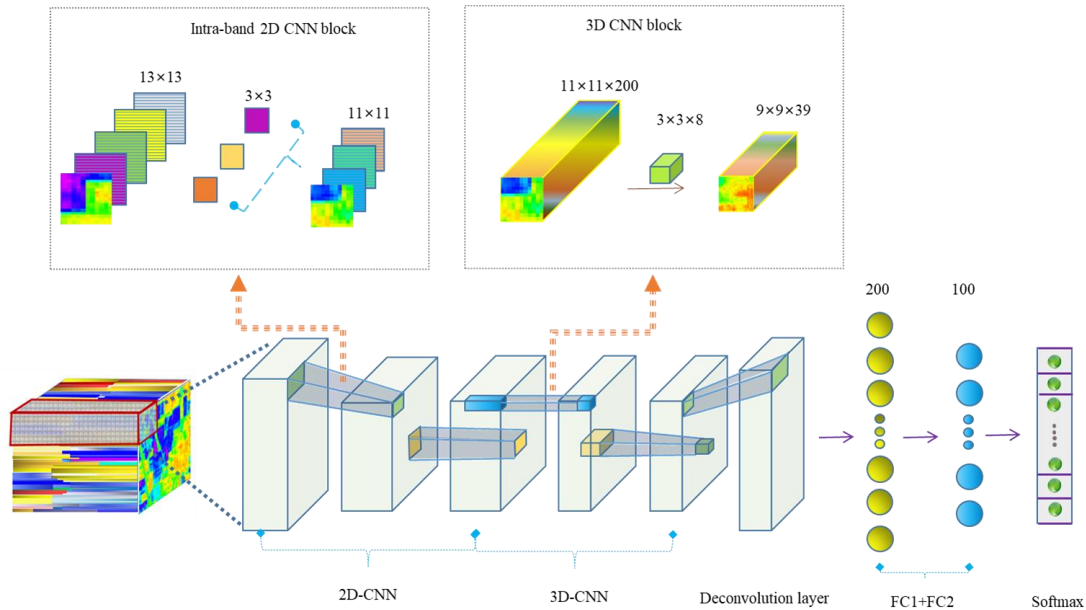


Fig. 1 Hierarchical structure diagram of the convolution neural network

B. 2D CNN with intra-band information

The training of the 2D CNN block includes two layers, the first layer is a 2D convolution layer to mine the given HSI data slice in a local perception mechanism. Assume that the sample of a hyperspectral image is denoted as X with dimension of $m \times n \times u$, here $X \in \mathbb{R}_0^{m \times n \times u}$ is a 3D tensor, $m \times n$ means the size of the training sample, u represents the

number of spectral bands, $X(i)$ denotes the spatial domain on the i th band of X . The spatial-spatial feature extraction using 2D-CNN is obtained by the following formula,

$$X_{2D} = C_{2D}(X(i)) = \frac{1}{u} \sum_{i=1}^u \sum_{j=1}^{t1} X(i) \Theta w_j^{2D} + b_j^{2D} \quad (1)$$

Here w_j^{2D} and b_j^{2D} represent the parameters and bias of the j th filter of size $a_1 \times b_1 \times t_1$, Θ means the 2D convolution operator.

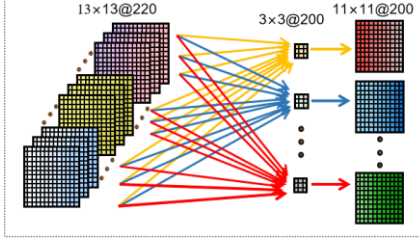


Fig. 2 Illustration of intra-band 2D CNN approach

The 2D convolution operator is usually applied to extract features in the spatial domain. To strength the feature representation ability, the spectral feature is utilized as channel information to participate in the production of feature extraction. Fig. 2 shows the illustration of the intra-band 2D CNN approach, we can observe that all the spectral bands are utilized to participate in the calculation of convolution, which can yield rich spatial features for the subsequent layer.

The second layer is called pooling layer, after the convolution layer, the nonlinear activation function is utilized as an activator to generate nonlinear feature in the 2D-CNN model, in this paper, we use zero-padding policy to keep the same size of next input feature for the following convolution layer, the specific equation of activator is listed as follow,

$$X_{2D}' = \max(0, \sum_{j=1}^{t_2} X_{2D} \Theta w_j' + b_j') \quad (2)$$

Here the size of the filter kernel is $a_2 \times b_2 \times t_2$, w_j' and b_j' represents the parameter and bias of the filter. In this module, the spatial feature is captured with the intra-band information involved, which is expressed in the feature quantification not only in the spatial region but also in the band domain.

C. 3D CNN for spectral feature fusion

The 3D CNN block is designed to complete spatial-spectral information fusion after the 2D CNN part. In this section, we consider the regular convolution layer for a 3D hyperspectral image represented by a 4D tensor $Y \in R_1^{n_h \times n_w \times l_h \times k_c}$ in the Tensorflow framework, where $n_h \times n_w \times l_h$ is the size of the input data, k_c denotes the number of the channel respectively, in this paper, we set the channel number k_c to 1 in the implementation framework. The refined feature through 3D CNN is denoted as Y_{3D} which is obtained by the 3D filter with the size $k_h \times k_w \times k_n$ applied to the input data by the following equation,

$$Y_{3D} = C_{3D}(X_{2D}') = \sum_{j=1}^N X_{2D}' \odot w_j^{3D} + b^{3D} \quad (3)$$

Here the symbol $\odot$ denotes the 3D convolution operator, w_j^{3D} and b^{3D} represent the parameter and bias of the 3D filter, N is the number of the 3D kernel. The 3D convolution operation is illustrated in Fig. 3, which is applied to a 3D block for the capture of the integration of spatial-spectral patch with a 3D kernel, the calculation of convolution is implemented in a sequential manner, e.g., from top to bottom, from left to right. Clearly, the complexity becomes progressively large as the parameters increase, which caused more training time and overfitting problems.

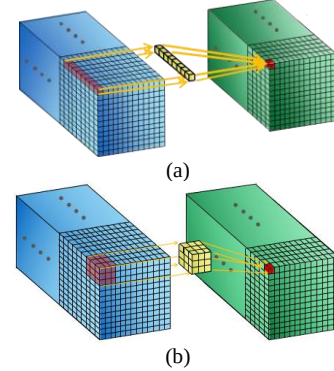


Fig. 3 Illustration of the 3D CNN. (a) with $1 \times 1 \times L$ ($L=8$) kernel, (b) with $3 \times 3 \times 3$ kernel

Different from the original 3D convolution, we designed a simplified 3D CNN with a filter kernel size of $1 \times 1 \times L$ to reduce approximately $k_h \times k_w$ times of the original parameters. According to the simplified structure, the 3D CNN block focuses on integrating the characteristic information of adjacent bands with the strike step. In this case, the 3D CNN model captured the spectral fusion features with faster speed. The key roles of the $1 \times 1 \times L$ kernel reflected in the two aspects, on one side, the implementation achieves an action to reduce the band dimension with a fusion way, on the other side, the operation performs exceptionally more nonlinear information for the feature extraction. Next, the activator function of *ReLU* to generate the nonlinear and sparse features of the CNN network is listed as follow,

$$Y_{3D}' = \max(0, Y_{3D} \odot w_{3D}' + b_{3D}') \quad (4)$$

Here w_{3D}' and b_{3D}' represents the parameter and bias of the pooling layer.

Compared with the regular CNN implementation, the proposed classification network for hyperspectral image consists of two steps separately, the 2D CNN performs spatial convolution independently including band information, followed by the 3D CNN convolution projecting the output of the spectral channels onto a new fusion feature to generate a powerful representation capability.

D. Remarks on the $1 \times 1 \times L$ kernel

In the 3D CNN part, it is worth noting that the 3D convolution operation with the kernel size fixed as $1 \times 1 \times L$ becomes to the 1D convolution conducted in 3-Dimensional space. In this section, the analysis of the 3D operation with $1 \times 1 \times L$ kernel is conducted in two aspects. On one hand, due to the fact that the traditional 1D CNN approaches for HSIC deal with the classification application in a pixel-wise way, the 1D CNN HSIC model usually needs to transform the sample training cube into a vector in a pixel-wise way firstly, and all the refined features have to combine to an integrated 3D patch after the convolution operation, which increased the time cost obviously with the data conversion back and forth. While with the $1 \times 1 \times L$ kernel, the 3D convolution operation can easily be done by the setting of the filters and the channel number. On the other hand, compared with the existing 1D-2D CNN architectures, the key to our proposed HSIC framework is to use 2D-1D networks to explicitly learn spectral and spatial feature jointly. Specific, the traditional 1D-2D CNN extracted the features from 1D CNN and 2D CNN in parallel part and the features are fused in a fully connection layer usually, which means that the implementation of feature extraction actually is produced separately. Instead of producing feature maps in independent, we employ the 2D CNN to extract abundant spatial features firstly and the subsequent 1D CNN block (3D with $1 \times 1 \times L$ kernel) aims to further refine the feature in the spectral domain. The proposed CNN attempts to perform the feature extraction with spectral and spatial information simultaneously to enhance the feature representation.

E. Optimization Details of the proposed CNN architecture

Furthermore, we adopted a series of optimization policy to improve the performance of the CNN Network in this paper. Firstly, in the preprocessing period, the mirroring strategy is used to enhance the classification accuracy of the pixels around the borders. The specific operation means to keep the border pixel be the center of the modified sample by expanding the original image. Secondly, we used the Root Mean Square prop (RMSprop) [37] as the optimization algorithm for the proposed network training, which is attenuated through a certain ratio by introducing an attenuation coefficient. Lastly, in the CNN network, the detailed structure of the feature is usually weakened or smoothed out, in order to enhance the expression of the extracted feature, we utilize deconvolution operation to expand the feature map as identical as the size of the input sample. A deconvolution operation aims to increase the size of the feature map, which is inverse to convolution operation and plays an important role in FCN and GAN networks to map low-resolution input into a high-resolution feature. The option can be implemented by upsampling to enlarge the

feature map with convolution kernel and padding, which has better effectiveness because it is a learning-based upsampling operation. For example, the size of input data is 5×5 , the extended feature map is 7×7 after zero-padding and the deconvolution procedure with a kernel size of 3×3 . In our paper, the deconvolution layer has a kernel filter with a size of 3×3 , and the length of the deconvolution filter is set to 39.

III. EXPERIMENT AND RESULT ANALYSIS

A. Data Description

In this paper, we have employed four well-known hyperspectral image data sets in the experiments.

1) Purdue Indiana Indian Pines Scene

The first one namely Purdue Indiana Indian Pines Scene is an AVIRIS image, which was collected over North-western Indiana. The image is characterized by 145×145 pixels with 220 spectral bands in the range of $0.4\text{--}2.5\mu\text{m}$. As shown in Fig. 2, the Purdue Indian Pines Scene consists of 16 classes available in the ground truth image.

2) Salinas Valley

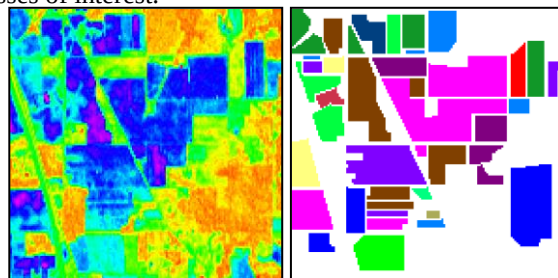
The second data set is the Salinas Valley scene captured by the AVIRIS sensor over the Salinas Valley in Southern California. The data contains 224 bands and the spatial resolution is 512×217 . According to the ground truth image shown in Fig. 3, there are 16 categories of classes labeled in different colors.

3) Kennedy Space Center (KSC)

The third data set used for the experiment is called KSC data, which was collected by AVIRIS in the range of $0.4\text{--}2.5\mu\text{m}$ of Kennedy Space Center located in Florida. After removing water absorption and low SNR bands, the sub-image with the size of 512×614 remains 176 bands for the analysis. The scene contains various land cover types represented by 13 classes labeled 1-13 as shown in Fig. 5.

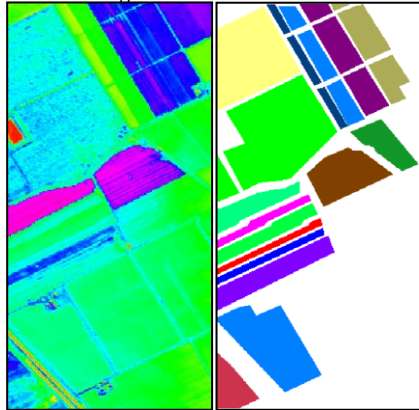
4) University of Pavia

The last data set used in the following experiment is the University of Pavia acquired by the ROSIS-03 satellite sensor. This scene has 103 spectral bands and 610×340 pixels with a spatial resolution of 1.3 m . Fig. 6 shows the Ground truth image of the University of Pavia with nine classes of interest.



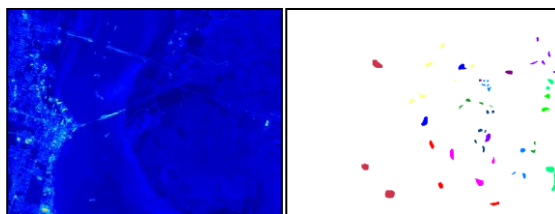
- | | |
|------------------|-----------------------------|
| 1. Alfalfa | 9. Grass-pasture-mowed |
| 2. Corn-notill | 10. Soybean-notill |
| 3. Corn-mintill | 11. Soybean-mintiall |
| 4. Corn | 12. Soybean-clean |
| 5. Grass-pasture | 13. Wheat |
| 6. Grass-trees | 14. Woods |
| 7. Oats | 15. Bldg-grass-trees-drives |
| 8. Hay-windrowed | 16. Stone-steel-towers |

Fig. 4 Ground truth image of Purdue Indiana Indian Pines Scene



- | | |
|----------------------|-----------------------------|
| 1. Weeds 1 | 9. Soil vineyard develop |
| 2. Weeds 2 | 10. Corn sensed green weeds |
| 3. Fallow | 11. Lettuce 4 weeks |
| 4. Fallow rough plow | 12. Lettuce 4 weeks |
| 5. Fallow smooth | 13. Lettuce 4 weeks |
| 6. Stubble | 14. Lettuce 4 weeks |
| 7. Celery | 15. Vineyard untrained |
| 8. Grapes untrained | 16. Vineyard trells |

Fig. 5 Ground truth image of Salinas valley



- | | |
|-----------------------------|-----------------------------|
| 1. Scrub | 8. Graminoid marsh |
| 2. Willow swamp | 9. Spartina marsh |
| 3. Cabbage palm hammock | 10. Corn sensed green weeds |
| 4. Cabbage palm/oak hammock | 11. Salt marsh |
| 5. Slash pine | 12. Mud flats |
| 6. Oak/broadleaf hammock | 13. Water |
| 7. Hardwood swamp | |

Fig. 6 Ground truth image of KSC



Fig. 7 Ground truth image of University of Pavia

B. Experimental Configuration

In this section, the hyperspectral image classification methods based on the three data sets were performed to evaluate the proposed CNN model. The experiments were run on a computer with Intel (R) Core (TM) i7-7820X CPU, 3.60Ghz, Nvidia Geforce GTX 1050Ti, RAM 32.0GB for the Salinas data, GTX 1060Ti, RAM 16.0GB for the other three data sets, the platform is python 3.6 of the Tensorflow framework. We evaluate the proposed architecture with classification performances in terms of Overall Accuracy (OA), here OA measures the number of samples is classified correctly.

We also exploited a series of other CNN model for comparison with the proposed architecture, here denoted the abbreviation symbols of the compared CNN networks. The network proposed in the paper is called as 2D-3D-D model, the 3D part with simplified kernel is denoted 2D-3D-D-S, the 2D CNN methods has only one 2D convolution layer, one pooling layer, and two fully connected layers, the 2D+D CNN network pluses one deconvolution layer and the 2D+3D CNN model include one 2D convolution layer, one 3D convolution layer, one pooling layer, and two fully connected layers. The hyperparameters setting of the proposed 2D-3D-D models are listed in Table 1. The other compared existing methods including SVM [38], EPF [39], LCMV [40], MFASR [41]. In addition, in our experiment for the CNN networks, the batch size is initialized to 100, the iteration number of training is fixed to 5000. All the other CNN networks mentioned for comparison have the same parameters as 2D-3D-D architecture. Each execution of all the CNN networks has been repeated 5 times and the classification accuracy reported in our experiment is averaged by the results, which is represented in the form of mean $\pm$ standard deviation.

Table 1 Hyperparameters setting of the proposed models

Type	2D-3D-D-S	2D-3D-D
Batch size	100	100
LR	0.001	0.001
2D CNN	3*3*200 stride=[1,1]	3*3* 200 stride=[1,1]

3D CNN	1*1*8*1 stride=[1,1,5]	3*3*3*1 stride=[1,1,1]
Deconvolution	3*3, 200 stride=[1,1]	3*3, 200 stride=[1,1]
FC 1	200	600
FC 2	100	150

Furthermore, it is noted that the existing challenging problems of HSIC are the few-shot learning and the unbalanced sample, therefore the data augmentation is usually a necessary operation to generate the samples before the training launch. In the preprocessing phase, we expand the training data set by augmentation in this paper. The specific types of addition are listed as follows: (1) Reverse the original data from up to down, (2) Reverse the training sample from left to right, (3) Increase data by adding random Gaussian noise to the training sample. For a specific class, we gather the data augmentation according to the number of the total samples of the ground truth image, specifically, when the number of total samples is less than 200, the percentage number of augmentation is set to 1/3 and the augmentation percentage is set to 1/5 for other situation. During the preprocessing phase, the training data is randomly selected from the original data sets and is expanded with the above ways randomly, where the mean value of Gaussian noise is set to 0 and the variance is 0.05. The specific numbers of the training sample of the three data sets are reported in table 2-5, which included the number selected randomly from every class and the number generated by the augmentation method, and the number in the column of *Augmentation* denotes the sum number of selected and extended sample

Table 2 Number of samples in the training set of the Purdue Indiana Indian Pines Scene used in 2D-3D-D CNN methods

Class No	Class Name	Sample Number			
		Total	Training	Augmentation	Testing
1	Alfalfa	46	5	20	41
2	Corn-notill	1428	143	428	1285
3	Corn-mintill	830	83	249	747
4	Corn	237	24	71	213
5	Grass-pasture	483	49	145	434
6	Grass-trees	730	73	219	657
7	Grass-pasture-mowed	28	3	12	25
8	Hay-windrowed	478	48	143	430
9	Oats	20	2	8	18
10	Soybean-notill	972	98	292	874
11	Soybean-mintill	2455	246	737	2209
12	Soybean-clean	593	60	178	533
13	Wheat	205	21	62	184
14	Woods	1265	127	380	1138
15	Buildings-Grass-Trees-Dr	386	39	116	347
16	Stone-Steel-Towers	93	10	41	83

Table 3 Number of samples in the training set of the Salinas Valley used in 2D-3D-D CNN methods

Class No	Class Name	Sample Number			
		Total	Training	Augmentation	Testing
1	Brocoli_green_weeds_1	2009	101	502	1908
2	Brocoli_green_weeds_2	3726	187	932	3539
3	Fallow	1976	99	494	1877
4	Fallow_rough_plow	1394	70	348	1324
5	Fallow_smooth	2678	134	669	2544
6	Stubble	3959	198	989	3761
7	Celery	3579	179	894	3400
8	Grapes_untrained	11271	564	2818	10707

9	Soil_vinyard_develop	6203	311	1551	5892
10	Corn_senesced_green_w	3278	164	819	3114
11	Lettuce_romaine_4wk	1068	54	267	1014
12	Lettuce_romaine_5wk	1927	97	482	1830
13	Lettuce_romaine_6wk	916	46	229	870
14	Lettuce_romaine_7wk	1070	54	268	1016
15	Vinyard_untrained	7268	364	1817	6904
16	Vinyard_vertical_trellis	1807	91	452	1716

Table 4 Number of samples in the training set of the Kennedy Space Center used in 2D-3D-D CNN methods

Class No.	Class Name	Sample Number			
		Total	Training	Augmentation	Testing
1	Scrub	761	77	229	684
2	Willow swamp	243	25	73	218
3	Cabbage palm hammock	256	26	77	230
4	Cabbage palm/oak hammock	252	26	76	226
5	Slash pine	161	17	70	144
6	Oak/broadleaf hammock	229	23	68	206
7	Hardwood swamp	105	11	46	94
8	Graminoid marsh	431	44	130	387
9	Spartina marsh	520	52	156	468
10	Cattail marsh	404	41	121	363
11	Salt marsh	419	42	125	377
12	Mud flats	503	51	151	452
13	Water	927	93	278	834

Table 5 Number of samples in the training set of University of Pavia used in 2D-3D-D methods

Class No.	Class Name	Sample Number			
		Total	Training	Augmentation	Testing
1	Asphalt	6631	332	1658	6299
2	Meadows	18649	933	4662	17716
3	Gravel	2099	105	524	1994
4	Trees	3064	154	766	2910
5	Painted metal sheets	1345	68	337	1277
6	Bare Soil	5029	252	1257	4777
7	Bitumen	1330	67	333	1263
8	Self-Blocking Bricks	3682	185	921	3497
9	Shadows	947	48	237	899

C. Results and Analysis

The training sample percentage of the Purdue Indiana Indian Pines Scene is set to 10% randomly, the dropout rate is initialized to 0.3 for 2D-3D-D-S and 0.8 for 2D-3D-D, the sample size is fixed as 13×13. Fig.8 illustrated the classification maps of the data set with each classifier mentioned above, the performance of the proposed 2D-3D-D CNN network is much better than the compared models. Moreover, the accuracy of each class is demonstrated individually in Table 6 and the OA of the Purdue data is shown in the last line. According to the listed value, it can be observed that our proposed 2D-3D-D-S model achieves the overall accuracy of 97.98%, and the second (97.49%) OA is implemented by the MFASR model. It is 0.49% and 0.88% higher than 2D+D and 2D+3D model, respectively, and the best OA is 98.33% with the 2D-3D-D framework. Compared with the OA 94.88% of the 2D model, it is obtained that the deconvolution layer and 3D convolution operations have an

important positive effect on the feature refined under the same conditions of other network structure parameters. Also, it also can be seen that the proposed method generates the best accuracy of class 4, and class 7. Besides, we can also observe that the accuracy of each class of the proposed model

is relatively high, it can be concluded that the overall classification performance is more stable in the proposed framework, and there is no case where the accuracy is much higher or lower for different classes.

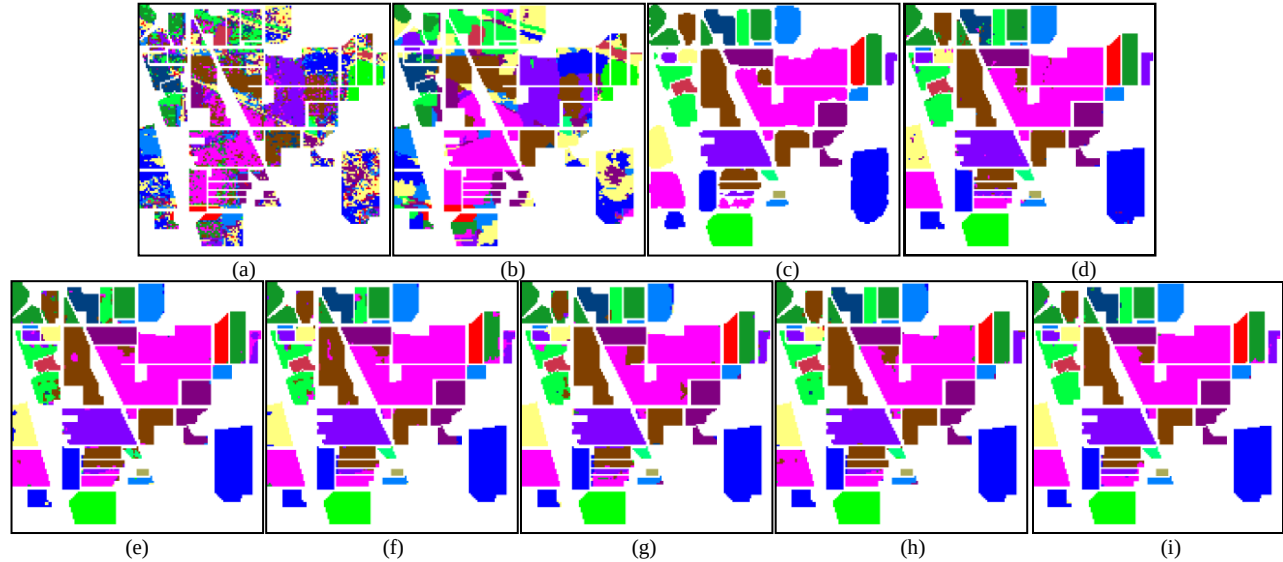


Fig. 8. The classification results of the Purdue Indiana Indian Pines Scene with compared methods. (a) SVM (b) EPF (c) LCMV (d) MFASR (e) 2DCNN (f) 2D+D (g) 2D+3D (h) 2D-3D-D-S (i) 2D-3D-D

Table 6 OA calculated from the classification results of Purdue Indiana Indian Pines Scene with all the compared methods (10%)

Class P _{OA} %	SVM	EPF	LCMV	MFASR	2D	2D+D	2D+3D	2D-3D-D-S	2D-3D-D
1	84.78	100	95.65	100	99.56±0.99	100.00±0.0	100.00±0.0	99.55±1.02	100±0
2	70.38	83.68	96.01	96.58	88.26±3.06	97.95±1.66	96.37±2.02	97.61±1.01	98.36±0.62
3	62.05	72.17	96.99	97.59	97.01±1.12	97.38±1.13	96.57±1.19	96.62±1.81	97.80±0.76
4	89.03	99.58	98.73	89.67	97.97±2.27	95.67±6.87	96.41±3.37	98.75±1.55	97.20±1.86
5	90.89	94.20	89.44	97.93	97.53±4.38	99.75±0.17	98.32±2.49	99.67±0.42	99.30±0.50
6	95.48	99.59	97.12	98.63	98.95±1.17	99.70±0.52	99.09±0.43	99.34±0.41	99.07±0.41
7	96.43	96.43	100	96.00	100.00±0.0	98.62±1.89	100.00±0.0	100±0	100±0
8	96.86	100	98.78	100	98.85±1.0	99.46±0.11	99.58±0.33	99.46±0.18	99.83±0.18
9	85.00	95.00	100	94.44	98.18±4.07	99.05±2.13	87.71±22.75	95.42±4.54	92.72±3.73
10	72.94	87.55	93.93	97.71	94.89±2.75	98.10±0.68	95.51±2.49	97.35±1.50	97.34±0.84
11	65.42	88.47	94.70	99.00	93.11±2.73	95.06±3.72	96.26±1.42	97.73±0.95	98.23±0.53
12	67.79	97.64	95.45	95.31	97.46±1.71	98.30±0.87	98.01±1.16	96.79±2.38	97.66±0.93
13	97.56	99.51	98.54	99.46	99.32±0.81	99.61±0.41	99.90±0.22	99.42±0.85	99.32±0.65
14	89.96	98.74	93.52	99.03	98.43±1.45	98.89±0.38	98.33±0.82	98.71±0.48	99.01±0.47
15	71.50	96.89	90.67	95.68	96.27±1.60	98.62±1.42	96.88±2.31	98.23±1.55	98.60±0.86
16	100	100	98.92	97.59	91.27±0.18	89.97±5.41	96.96±3.28	94.92±3.19	92.59±3.08
POA	76.47	90.78	95.09	97.83	94.88±0.89	97.49±1.03	97.10±0.51	97.98±0.19	98.33±0.21

The colorful classification maps of each method for the Salinas Valley are illustrated in Fig. 9, in this experiment, the training sample percentage is set to 5%, the setting of the dropout rate and the size of the training data is 0.3 for 2D-3D-D-S and 0.8 for 2D-3D-D and 13×13 respectively. Table 7 lists the accuracy of each class and OA of the Salinas data. Due to the Salinas data has a rich sample size and relatively regular spatial distribution, several network structures generate good performance with almost the same

overall accuracy. Objectively, it is clearly to be seen that the 2D-3D-D model obtained the best experimental results with an overall accuracy of 99.07%. In contrast, both the MFASR model and the 2D+D model showed good results with an overall accuracy of 97.91% and 97.90%, respectively. Also, our proposed model is relatively 0.3% higher than the 2D+3D model. According to the obtained value of the different classes, the proposed network can obtain a more stable accuracy.

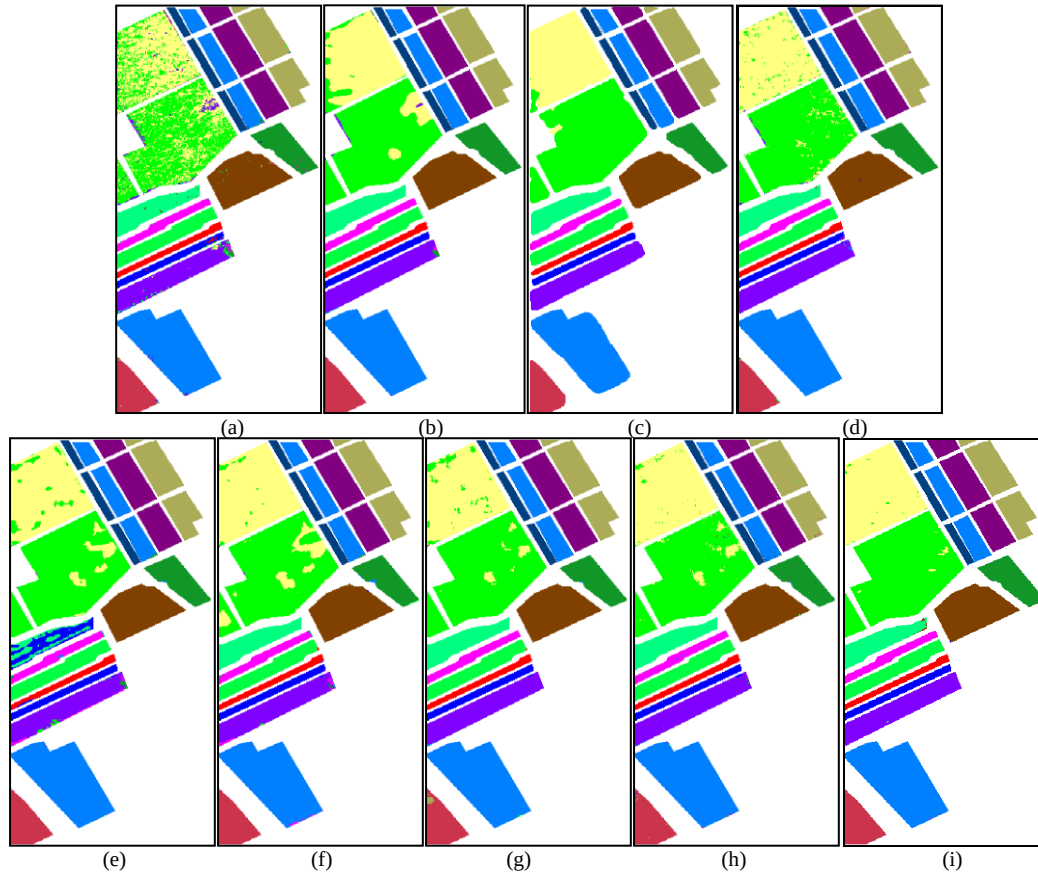


Fig. 9. The classification results of the Salinas valley with compared methods. (a) SVM (b) EPF (c) LCMV (d) MFASR (e) 2DCNN (f) 2D+D (g) 2D+3D (h) 2D-3D-D-S (i) 2D-3D-D

Table 7 OA calculated from the classification results of Salinas Valley with all the compared methods (5%)

Class P _{OA} %	SVM	EPF	LCMV	MFASR	2D	2D+D	2D+3D	2D-3D-D-S	2D-3D-D
1	99.15	100	95.52	99.90	99.91±0.20	95.78±8.53	99.98±0.04	99.43±0.28	99.81±0.42
2	99.30	100	98.42	99.66	93.15±4.83	97.03±6.64	99.70±0.58	99.69±0.50	99.65±0.36
3	98.94	100	93.78	99.95	93.75±7.22	99.55±0.70	96.41±4.97	99.50±0.47	99.75±0.22
4	99.43	100	95.62	99.85	99.61±0.43	99.17±1.08	99.27±0.98	99.06±0.63	99.37±0.68
5	97.80	99.33	96.90	99.17	99.04±0.70	98.38±3.13	99.78±0.21	99.25±0.32	98.68±1.97
6	99.75	100	98.79	99.57	99.78±0.16	99.99±0.01	100.00±0.00	99.87±0.06	99.99±0.02
7	99.75	100	98.63	99.24	95.14±8.77	99.74±0.30	99.90±0.18	99.60±0.50	99.88±0.13
8	71.91	91.08	96.69	94.95	85.40±3.51	95.37±2.70	95.17±3.39	97.17±1.46	98.05±1.13
9	99.21	99.95	95.87	100	99.80±0.26	99.91±0.07	99.47±0.46	99.82±0.17	99.80±0.19
10	91.06	97.47	96.67	99.00	99.63±0.24	99.92±0.18	99.92±0.05	98.85±0.83	99.86±0.08
11	98.78	100	97.75	99.70	76.60±11.25	99.31±0.45	99.83±0.21	96.61±4.14	98.67±0.98
12	99.84	100	97.15	100	93.77±5.15	99.96±0.09	99.96±0.07	98.83±1.83	99.92±0.13
13	99.24	100	96.51	97.59	99.69±0.27	99.96±0.06	99.96±0.10	98.96±0.82	99.89±0.19
14	93.74	99.91	95.89	97.05	84.06±21.13	99.43±0.68	98.62±1.38	98.72±3.68	99.40±1.00
15	70.65	89.97	94.00	94.93	94.36±3.97	96.43±2.21	93.94±5.48	93.52±4.34	97.76±1.82
16	98.56	100	93.30	98.78	99.96±0.06	100.00±0.00	99.96±0.10	99.72±0.22	99.88±0.19
POA	89.09	96.60	96.37	97.91	92.24±5.20	97.90±1.09	97.77±0.45	98.07±0.60	99.07±0.38

In the experiment for the KSC data, we fix the training sample percentage as 10%, and the data size and the dropout rate are set to 0.3, the kernel numbers of FC1 and FC2 for 2D-3D-D are 600 and 150. Fig. 10 demonstrates the classification results with the compared methods. The accuracy of each class and OA are illustrated in Table 8 objectively, it can be seen from the table that our CNN architecture yielded an overall accuracy of 97.14% and

97.47%. The MFASR model based on sparsity generates the results of 98.31%, followed by 2D+D and 2D+3D models with the OA of 96.30% and 93.58%, respectively. Due to the super sparsity of the KSC data itself, the MFASR has the better experimental result than our proposed method, although, for the class 5, the OA is only 79.86% in MFASR, while the OA of our proposed network is 84.59% and 94.10%, it can be analyzed that our network still has more stable

performance for each class than other classifiers in the KSC data.

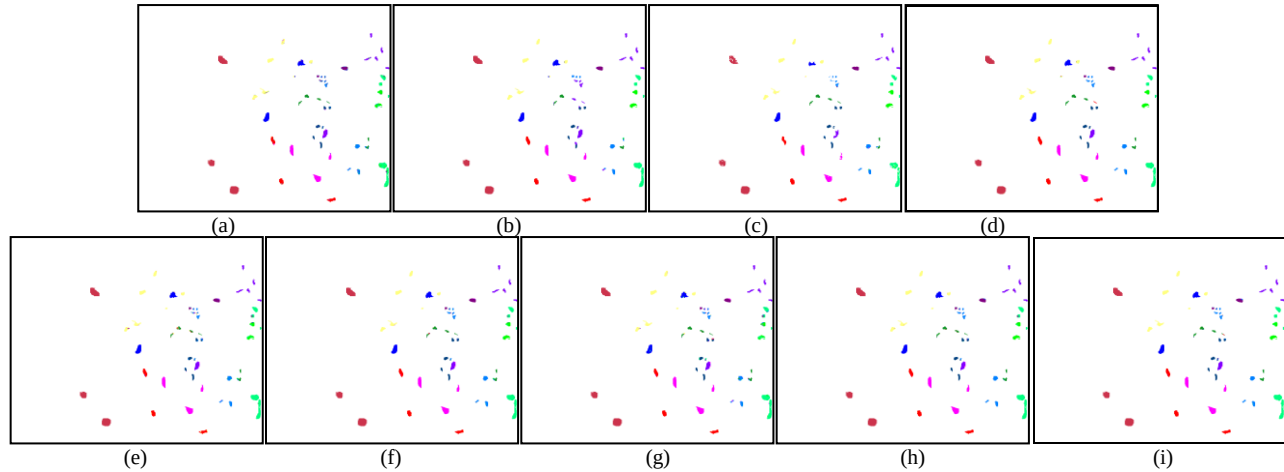


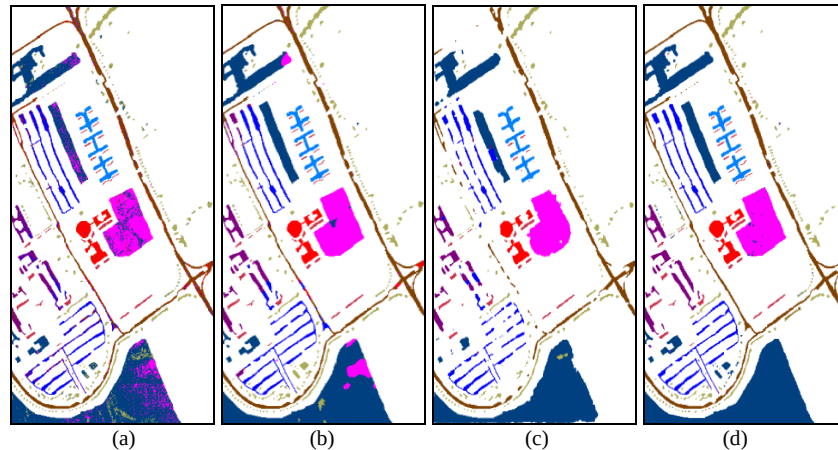
Fig. 10. The classification results of the KSC data with compared methods. (a) SVM (b) EPF (c) LCMV (d) MFASR (e) 2DCNN (f) 2D+D (g) 2D+3D (h) 2D-3D-D-S (i) 2D-3D-D

Table 8 OA calculated from the classification results of Kennedy Space Center with all the compared methods (10%)

Class P _{OA} %	SVM	EPF	LCMV	MFASR	2D	2D+D	2D+3D	2D-3D-D-S	2D-3D-D
1	86.07	100	87.68	97.37	93.19±4.43	95.16±1.78	89.49±3.35	97.06±1.06	98.19±0.70
2	89.71	66.26	88.52	100	96.38±1.74	97.54±1.84	92.66±2.94	95.52±2.43	96.75±2.31
3	91.41	80.47	93.02	100	97.88±1.36	97.19±1.66	96.04±2.67	97.33±1.73	98.31±0.32
4	75.00	53.17	80.24	99.12	82.01±7.46	90.34±5.89	87.83±2.50	89.79±4.14	87.50±1.24
5	83.85	80.12	79.50	79.86	85.06±4.49	87.51±3.85	83.60±3.27	84.59±2.43	94.10±3.22
6	80.35	100	95.20	100	88.17±4.63	99.05±0.74	87.97±6.17	99.61±0.55	98.73±1.19
7	95.24	100	96.33	100	90.36±4.44	93.59±2.35	83.86±8.44	94.82±3.27	95.87±4.33
8	90.95	100	82.60	100	94.63±6.71	98.52±1.37	97.36±2.51	98.29±2.04	96.32±3.75
9	97.69	100	94.23	100	95.00±4.22	93.66±1.36	91.27±3.95	97.44±0.82	97.34±0.84
10	89.11	99.75	91.85	96.42	95.12±2.67	98.19±1.52	98.38±0.97	97.89±1.08	97.35±2.04
11	97.37	100	82.38	100	98.60±1.10	98.96±0.36	97.03±2.80	99.88±0.20	99.48±0.61
12	94.43	97.81	95.83	97.35	86.64±2.69	96.41±1.80	94.65±1.60	97.19±1.72	98.90±1.22
13	99.89	100	91.37	99.40	96.88±2.58	98.49±1.73	98.37±1.83	99.55±0.24	99.70±0.47
POA	91.81	94.36	89.45	98.31	93.04±0.86	96.30±0.41	93.58±1.58	97.14±0.26	97.47±0.39

For the experiment of the University of Pavia, the training sample percentage is 5%, the training size of the data set is fixed as 13×13, and the drop out rate is set to 0.3 for 2D-3D-D-S and 0.8 for 2D-3D-D. Fig. 11 illustrates the classification results with the mentioned HSIC methods, the accuracy of each class and OA of Pavia data is shown in

Table 9 objectively. As can be observed, the 2D-3D-D model wins the other compared classification methods in terms of the OA criterion, the proposed architecture achieves the best OA of 99.54% and the highest OA of class No. 2, 4, 5 and 9 especially, which are 99.93%, 99.88%, 99.97%, and 99.96%.



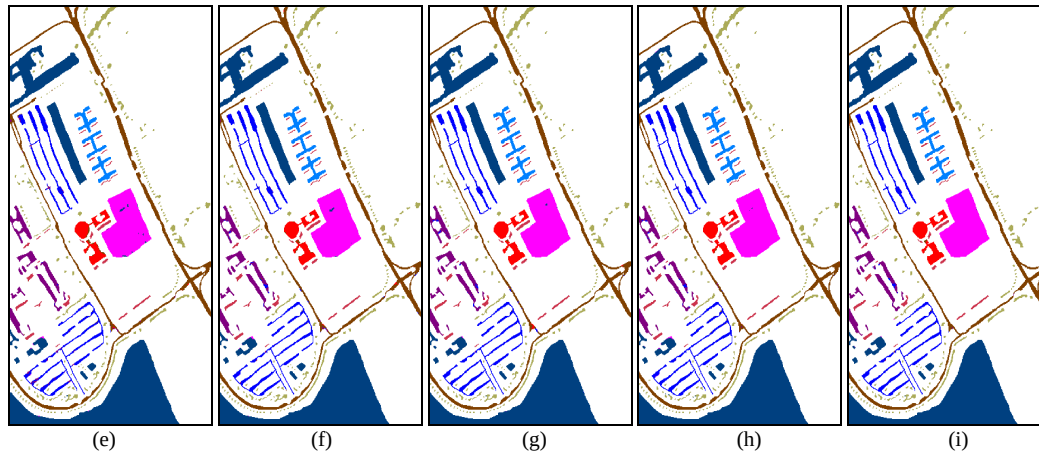


Fig. 11. The classification results of the University of Pavia with compared methods. (a) SVM (b) EPF (c) LCMV (d) MFASR (e) 2DCNN (f) 2D+D (g) 2D+3D (h) 2D-3D-D-S (i) 2D-3D-D

Table 9 OA calculated from the classification results of University of Pavia with all the compared methods (5%)

Class P _{OA} %	SVM	EPF	LCMV	MFASR	2D	2D+D	2D+3D	2D-3D-D-S	2D-3D-D
1	93.75	90.47	77.24	99.68	98.86±0.33	98.90±0.29	99.45±0.08	98.79±0.90	99.42±0.05
2	94.03	91.95	86.42	99.90	99.49±0.43	99.89±0.03	99.92±0.10	99.91±0.06	99.93±0.24
3	67.41	87.18	73.34	99.70	97.57±1.41	96.98±0.88	97.67±1.51	97.60±1.96	98.69±0.32
4	76.52	98.37	79.85	90.24	99.50±0.81	99.86±0.05	99.77±0.31	99.81±0.18	99.88±0.16
5	97.68	99.93	98.81	99.61	99.96±0.03	99.88±0.16	99.91±0.09	99.75±0.20	99.97±0.04
6	62.90	97.93	91.49	98.12	98.58±1.46	99.26±0.11	99.55±0.25	99.18±0.20	99.45±0.32
7	57.29	100	89.10	99.76	95.73±3.05	97.38±1.66	98.12±1.42	99.47±0.32	99.47±0.45
8	83.47	90.77	81.10	99.20	96.70±0.87	97.45±0.53	97.47±0.67	97.48±0.67	97.89±0.58
9	99.86	100	78.46	99.11	99.79±0.15	99.50±0.54	99.75±0.14	99.68±0.31	99.96±0.09
POA	84.13	93.23	84.32	98.87	98.87±0.22	99.22±0.04	99.41±0.14	99.29±0.27	99.54±0.11

Since the parameters play a significant role in the 3D part of the model, for fair comparison with the same number of neurons of fully connection layers, a series of experiments are conducted for all the four data sets to show the difference of the proposed 2D-3D-D-S and 2D-3D-D methods. All the classification results by the average of 5-times execution are listed in Table 10, where the “FC-A” denotes the number of FC1 and FC2 is 200 and 150, the “FC-B” describes the circumstance with the number of FC1 and FC2 is 600 and 150 respectively. In the last row, the OA with FC-A is illustrated in the color of light orange and the OA with FC-B is performed in the color of light green. It can be observed that the 2D-3D-D-S has better

performance with all the data sets in the situation of FC-A, while 2D-3D-D is more outstanding in the FC-B environment. Due to the fact listed below, all the results are reasonable and plausible. For one thing, the 2D-3D-D adopted to learn the feature with numerous parameters which is disadvantage to the classification performance in the fewer kernels in the fully connection layers. For the other thing, the complex model performs better reveals that 2D-3D-D extracted more efficient feature maps with more parameters. That is to say, the 2D-3D-D-S in FC-A outperform the 2D-3D-D model, while the 2D-3D-D model is more challenging in the condition of the FC-B.

Table 10 OA calculated of the four data sets with the proposed method with different FC parameters

Class P _{OA} %	Purdue				Salinas				KSC				Pavia			
	2D-3D-D-s		2D-3D-D		2D-3D-D-s		2D-3D-D		2D-3D-D-s		2D-3D-D		2D-3D-D-s		2D-3D-D	
	FC-A	FC-B	FC-A	FC-B	FC-A	FC-B	FC-A	FC-B	FC-A	FC-B	FC-A	FC-B	FC-A	FC-B	FC-A	FC-B
1	99.55	96.96	98.26	100	99.43	93.13	98.24	99.81	97.06	99.33	99.01	98.19	98.79	98.06	97.78	99.42
2	97.61	93.36	95.90	98.36	99.69	99.78	98.99	99.65	95.52	92.33	95.30	96.75	99.91	99.90	99.94	99.93
3	96.62	96.46	94.07	97.80	99.50	99.89	99.55	99.75	97.33	97.70	97.04	98.31	97.60	97.08	94.44	98.69
4	98.75	98.06	98.14	97.20	99.06	99.64	99.49	99.37	89.79	92.91	91.29	87.50	99.81	99.03	98.28	99.88
5	99.67	94.08	94.66	99.30	99.25	96.97	99.34	98.68	84.59	94.24	93.33	94.10	99.75	99.99	99.96	99.97
6	99.34	99.07	99.43	99.07	99.87	100	99.99	99.99	99.61	98.60	99.35	98.73	99.18	99.64	99.54	99.45
7	100	97.14	95.00	100	99.60	99.77	99.62	99.88	94.82	99.09	97.95	95.87	99.47	98.81	97.14	99.47
8	99.46	100	100	99.83	97.17	97.80	95.57	98.05	98.29	93.85	95.96	96.32	97.48	98.04	98.27	97.89
9	95.42	90.00	98.00	92.72	99.82	99.97	99.56	99.80	97.44	94.90	95.42	97.34	99.68	99.93	99.64	99.96

10	97.35	97.41	98.29	97.34	98.85	99.55	99.55	99.86	97.89	95.43	96.00	97.35	-	-	-	-
11	97.73	97.35	98.57	98.23	96.61	99.76	99.94	98.67	99.88	97.87	98.72	99.48	-	-	-	-
12	96.79	92.58	92.92	97.66	98.83	99.74	99.22	99.92	97.19	96.73	96.69	98.90	-	-	-	-
13	99.42	99.22	99.41	99.32	98.96	99.67	99.72	99.89	99.55	99.51	99.30	99.70	-	-	-	-
14	98.71	99.70	99.90	99.01	98.72	99.42	99.68	99.40	-	-	-	-	-	-	-	-
15	98.23	98.45	99.53	98.60	93.52	91.32	92.30	97.76	-	-	-	-	-	-	-	-
16	94.92	100	98.28	92.59	99.72	99.62	98.40	99.88	-	-	-	-	-	-	-	-
POA	97.98	96.94	97.62	98.33	98.07	97.85	97.65	99.07	97.14	96.77	97.04	97.47	99.29	99.20	98.94	99.54

In order to fully demonstrate all the aspects of the proposed network, we also recorded the convergence curves and the changing trend of the training loss of the four data sets with the above parameters. The loss curve is shown in Fig. 12, respectively, the vertical axis reflects the evolution of the error, the number of the iteration is listed on the horizontal axis, and the shadow represents the standard deviation of the curve fitting. It can be concluded there is a very little difference in the curve between the Purdue Indiana Indian Pines Scene and the KSC data, the loss values tend to be zero after 2000 iteration demonstrates that the proposed CNN model converges well, and the reasonable and similar training loss curves also show that our method is effective with the RMSprop optimization. As mentioned, we chose 5000 times as the final number of training iterations to obtain better precision results in the experiments. In the training phase, the training loss is feedback to the machine for back propagation, which reflects the training state of the network. The 2D-3D-D model used 3D kernel for the spectral-spatial feature extraction, which generated more spatial redundancy than the 2D-3D-D-S model especially for the sparse data. Due to fact that the KSC data is too sparse as shown in the ground truth image, it is more sensitive to the feature representation ability, therefore, the convergence of the curve of KSC data is slower than other data sets as shown in Fig. 12 (b) for the redundancy caused by the 2D-3D-D architecture.

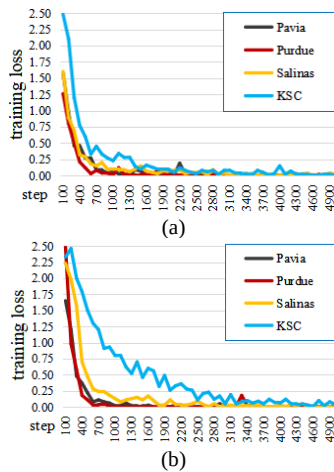


Fig.12 The training loss of the proposed CNN framework. (a) 2D-3D-D-S (b) 2D-3D-D

The training time of the proposed model with a comparison of the other CNN networks on the three

hyperspectral data sets is listed in Table 11. The computational cost is related to the size of the image, the numbers of the total layers, and the training sample, we can observe that CNND [42] has the longest time cost for the Salinas data, while the Purdue data has the smallest training time for our proposed framework. Also, it can be seen that with the same 7 layers, our proposed 2D-3D-D-S model is quicker than the CNN model for the Purdue data, Pavia data, and Salinas data, and has quite the same cost on the KSC data, the 2D-3D-D network has more cost than the 2D-3D-D-S due to the 3D kernel in 2D-3D-D is more complex than the 2D-3D-D-S model.

Table 11 The training time of the above CNN frameworks (Seconds)

Time (Seconds)	CNNS (4 layers)	CNNT (6layers)	CNND (7 layers)	2D-3D-D-S (7 layers)	2D-3D-D (7 layers)
Purdue	363.20	465.60	671.87	317.19	505.88
Salinas	679.16	786.44	995.03	964.67	1497.08
KSC	396.22	430.92	654.7	657.67	1288.15
Pavia	354.58	415.78	632.74	455.23	807.06

D. Comparison with the state-in-art CNN architectures

To demonstrate the effectiveness of the proposed framework, we compared the 2D-3D-D model with the other five state-of-the-art CNN models on the Purdue Indiana Indian Pines Scene. The approaches included in the comparison are summarized as follows. CNN-MRF model [22] adopted CNN to learn the posterior class distributions for better spatial information extraction, which included two convolution layers with the kernel size of 5×5 and 3×3 , the numbers of the kernel of two fully connected layers are set to 200 and 100. The learning rate is 0.001 and the batch size is set as 100. The SS-3DCNN [29] is a semi-supervised 3-D CNN for deep feature extraction, in the model, the size of the input sample is $25 \times 25 \times 12$, the learning rate is 0.005 and the training epoch is set to 150, respectively. In SRCL [21], the author proposed deep network architecture for a super-resolution aided HSIC with class-wise loss, the size of the convolution kernel is set to 9×9 , 1×1 , and 5×5 , respectively, the filter size of the pooling layer is fixed to 2×2 . In [27], the 3D auto model, the author designed 1-D Auto-CNN and 3-D Auto-CNN for spectral and spectral-spatial HSI classifiers, 3×3 and 5×5 separable convolution, 3×3 and 5×5 dilated separable convolution, 3×3 average pooling and 3×3 max-pooling are used in the 3D auto model. The training epoch for the 1-D Auto-CNN and 3-D Auto-CNN-Cutout is set to 300 and 100. In 3D SP DNN [26], the model exploits the 3D spectral-spatial information via super pixel-based neural networks. The sizes

of filter are set $4 \times 4 \times 63$ and $3 \times 3 \times 62$, the numbers of the filter are set to 6 and 12, the kernel size is set to $2 \times 2 \times 2$ for the max-pooling process. The 3D-CNN [43] model is composed of two 3D convolution blocks (C1 and C2) followed by a fully connection layer (F1), the size of the 3D convolutional kernel of C1 is $3 \times 3 \times 7$, the number is set to 2, and the size of kernel is $3 \times 3 \times 3$, the number of filters is 4 for C2, respectively. FLCNN [44] is designed to learn sensor-specific spatial-spectral features with 5-layers CNN, which can be implemented the HSIC in both unsupervised and supervised way, the size of input sample is 3×3 and 5×5 , the number of kernel is 20 in the convolution layer and the number of sigmoid neuron nodes in the fully connection layer is set to 100. A 3-DCNN network [45] captured the spatial and spectral context, which is consisted of seven convolution layers with 3D kernel and one full connection layer, the kernel sizes of the 3D operation included $3 \times 3 \times 3$, $1 \times 1 \times 3$ and $1 \times 1 \times 2$ specifically. In [46], the DC-CNN utilized two-dimensional CNN the hierarchical feature extraction, the 1D CNN channel and 2D channel extract the features with the 1D and 2D convolution operation. The 1D unit contains two pairs of convolutional layers, it has 300 and two convolutional kernels for the sequential convolutional layer, the 2D-CNN contains two pairs of convolutional layers, the size of the convolutional layer is fixed to 3×3 . Table 12 to 15 expressed the comparison of classification accuracy obtained by the CNN models for comparison, we compare the detailed

classification performance on each class for the four data sets. As can be seen, for the Purdue Indiana Indian Pines Scene data, the 3-D-CNN method achieves the highest classification OA than the other models, however, we investigate that the 3-D-CNN acquired OA of 99.07% with the training percentage of 70%, while the proposed 2D-3D-D yields OA of 98.33% with only 10% training sample. For fair comparison, we have done the experiment on the KSC data set with the same ratio, the classification results are listed in last column in Table 15, it can be seen that the better performance of our model which generates the OA of 99.95%. For the Salinas valley and the University of Pavia, the proposed framework generates the best classification performance as shown in Table 13 and 14, and Table 15 shows that the proposed CNN almost get the best OA with 40% training sample for the Kennedy Space Center. Furthermore, to evaluate the performance of our network objectively, we also have done a series of the experiment of the classification without sample augmentation for all the four data sets, each execution of the CNN network have been repeated 3 times in this part, and we reported the accuracy averaged by the results in Table 16. It is easy to be observed that the 2D-3D-D is effective without data expansion and the OA is stable for all the data sets. To sum up, from the above results and analysis, we can conclude that the proposed architecture achieves better classification performance compared with state-of-the-art approaches.

Table 12 OA calculated from the classification results of the Purdue Indiana Indian Pines Scene with the compared methods

Class P _{OA} %	CNN-MR F (40 per class)	3-D Auto-CN N-Cutout (5%)	SRCL (98%)	SS-3DCNN (2%)	3D SP-DNNs (1%)	3D-CNN (50%)	FL-CNN (200 per class)	3-D-CNN (70%)	DC-CNN (2 per class)	2D-3D-D-S (10%)	2D-3D-D (10%)
1	90.24	33.14	89.58	97.96	100	95.89	80.43	85.71	67.44	99.55	100
2	96.96	88.28	97.62	96.49	91.11	98.46	95.92	96.46	96.39	97.61	98.36
3	98.26	79.86	97.78	99.53	85.78	98.99	98.22	97.13	98.10	96.62	97.80
4	96.71	56.57	81.91	97.47	24.89	99.14	93.34	98.55	97.78	98.75	97.20
5	97.24	71.13	98.51	97.21	81.57	99.29	97.52	97.90	94.32	99.67	99.30
6	99.24	95.58	98.76	95.24	89.59	99.92	98.89	97.68	98.85	99.34	99.07
7	92.00	48.18	71.02	98.88	100	100	95.63	100	100	100	100
8	99.77	99.77	100.0	98.51	66.32	100	99.86	99.30	100	99.46	99.83
9	0	20.00	3.7	97.73	100	92.31	68.33	100	100	95.42	92.72
10	98.40	91.74	98.55	94.44	77.88	98.12	95.66	98.26	90.25	97.35	97.34
11	98.05	93.70	98.54	97.83	88.43	98.96	97.47	98.77	97.04	97.73	98.23
12	98.31	73.70	94.57	97.70	66.95	98.99	97.86	97.15	92.18	96.79	97.66
13	100	94.75	99.65	97.24	99.51	99.82	98.76	96.72	98.45	99.42	99.32
14	96.49	98.20	99.00	96.47	99.37	99.81	98.46	99.46	99.50	98.71	99.01
15	100	50.99	98.74	95.81	83.68	99.56	89.21	93.80	98.63	98.23	98.60
16	86.75	70.57	94.12	99.83	98.92	99.38	98.21	100	97.73	94.92	92.59
POA	97.10	89.01	97.58	97.89	85.20	99.07	96.99	97.31	96.55	97.98	98.33
AA	90.53	72.88	88.88	97.39	84.63	98.66	93.99	98.92	95.42	98.09	97.94

Table 13 OA calculated from the classification results of the Pavia university with the compared methods

Class P _{OA} %	CNN-MR F (40 per class)	3-D Auto-CN N-Cutout (5%)	SRCL (98%)	SS-3DCNN (2%)	3D SP-DNNs (1%)	3D-CNN (50%)	FL-CNN (200 per class)	3-D-CNN (70%)	DC-CNN (2 per class)	2D-3D-D-S (5%)	2D-3D-D (5%)
1	98.02	94.14	97.37	98.01	97.50	99.65	97.40	98.70	97.51	98.79	99.42
2	97.78	92.78	99.90	98.20	99.78	99.83	99.40	99.77	99.44	99.91	99.93
3	88.47	80.60	86.24	98.15	90.27	94.65	94.84	97.94	92.08	97.60	98.69
4	99.17	83.42	98.15	99.23	90.07	99.09	99.16	94.55	98.17	99.81	99.88
5	99.90	99.13	100	99.31	99.74	100	100	97.77	99.63	99.75	99.97
6	93.00	95.62	98.34	99.41	99.73	99.93	98.70	99.60	98.37	99.18	99.45

7	87.47	87.31	81.29	98.92	99.76	97.75	100	98.00	91.49	99.47	99.47
8	91.66	98.39	90.40	98.08	100	99.24	94.57	98.55	95.76	97.48	97.89
9	98.03	63.02	99.30	98.06	99.29	99.55	99.87	81.34	99.26	99.68	99.96
POA	96.18	93.88	97.12	98.45	98.47	99.39	98.41	98.49	98.00	99.29	99.54
AA	94.83	88.19	94.55	98.60	97.35	98.85	98.22	96.25	97.35	99.10	99.41

Table 14 OA calculated from the classification results of the Salinas valley with the compared methods

Class P _{OA} %	3-D Auto-CNN -Cutout (4%)	SRCL (98%)	SS-3DCNN (10%)	FL-CNN (10%)	3-D-CNN (70%)	2D-3D-D-S (5%)	2D-3D-D (5%)
1	99.24	98.67	96.73	100	97.68	99.43	99.81
2	91.20	98.30	98.50	99.89	99.46	99.69	99.65
3	94.19	97.53	96.06	99.89	97.80	99.50	99.75
4	92.38	98.09	98.80	99.25	97.13	99.06	99.37
5	99.76	97.55	97.88	99.39	98.80	99.25	98.68
6	95.57	98.93	98.87	100	98.91	99.87	99.99
7	100	97.33	96.58	99.82	96.55	99.60	99.88
8	96.13	97.16	98.61	91.45	97.28	97.17	98.05
9	100	99.00	98.92	99.95	99.84	99.82	99.80
10	100	96.68	98.30	98.51	98.78	98.85	99.86
11	100	96.98	98.96	99.31	99.06	96.61	98.67
12	96.43	99.60	99.71	100	99.14	98.83	99.92
13	100	97.82	98.78	99.72	90.55	98.96	99.89
14	97.61	97.51	98.96	100	93.46	98.72	99.40
15	97.29	94.48	98.01	96.24	97.47	93.52	97.76
16	99.24	97.17	98.77	99.63	99.06	99.72	99.88
POA	91.20	97.42	98.29	97.42	99.08	98.07	99.07
AA	94.19	97.67	98.28	98.94	98.65	98.66	99.37

Table 15 OA calculated from the classification results of the KSC dataset with the compared methods

Class P _{OA} %	3-D Auto-CNN -Cutout (40%)	3-D-CNN (70%)	2D-3D-D-S	2D-3D-D (10%)	2D-3D-D (40%)	2D-3D-D (70%)
1	99.24	98.68	97.06	98.19	98.77	100
2	91.20	91.78	95.52	96.75	98.47	100
3	94.19	98.78	97.33	98.31	97.02	100
4	92.38	97.37	89.79	87.50	96.88	99.34
5	99.76	91.67	84.59	94.10	97.28	99.59
6	95.57	97.10	99.61	98.73	98.96	100
7	100	100	94.82	95.87	99.76	100
8	96.13	100	98.29	96.32	95.54	100
9	100	99.36	97.44	97.34	98.38	100
10	100	94.12	97.89	97.35	98.54	100
11	100	100	99.88	99.48	99.11	100
12	96.43	96.69	97.19	98.90	98.67	100
13	100	97.12	99.55	99.70	99.81	100
POA	97.61	98.46	97.14	97.47	98.42	99.95
AA	97.29	98.20	96.07	96.81	98.26	99.92

Table 16 OA calculated from the classification results of the four data sets with the 2D-3D-D method without augmentation

Data sets	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	OA
Purdue	87.03	92.10	91.40	90.13	96.13	95.00	90.39	98.65	91.97	92.50	95.56	90.42	98.67	97.00	93.55	96.55	94.26
Salinas	100	99.97	99.93	99.52	99.58	99.97	99.89	98.22	100	99.10	97.90	99.95	99.81	99.69	95.68	99.56	98.86
KSC	96.12	93.49	88.55	85.92	98.40	99.19	94.56	88.15	97.16	97.68	99.61	96.98	98.55	—	—	—	95.48
Pavia	98.41	99.75	95.49	99.42	99.80	99.29	98.39	93.47	97.79	—	—	—	—	—	—	—	98.61

E. Comparison of the different size of the training sample

In this part, the impact with different training size of the three data sets is evaluated with the proposed 2D-3D-D CNN network. For the Purdue Indiana Indian Pines Scene and the KSC training sets, the patch sizes of input data are set to 9×9, 11×11, 13×13, 15 ×15, 17×17, 19×19 respectively, and we select 9×9, 10×10, 11×11, 12×12, 13×13, 14×14 as the size of the input sample for the Salinas data set and the Pavia data. The dropout rate for all the three data sets is fixed to 0.3, and the

percentage of the Purdue and KSC data is set to 10%, and the portion for the Salinas data is 5%, Fig. 13. shows the OA expression with the histogram obtained by our proposed network on three data sets. It can be observed that OA varies as the size of the training sample varies. The accuracy of the three data sets generally shows an increasing trend with the size of the space increases, among them, the KSC data set has the most obvious promotion with the size is enlarged. As shown in Fig.13 (a), it can be observed that the best results are generated with the size of 19×19 for the Purdue and KSC data sets in the

two 2D-3D-D models. When the sample size is set to 9×9 , the lowest value of OA is 94.53% for the Purdue data with the simplified kernel, 96.02% for the Salinas data, and 92.79% for the KSC data separately. For the 2D-3D-D model, the lowest OA value is 97.15%, 97.08% and 91.12% for the Purdue data, the Salinas data and the KSC data with the smallest size of the training patch, while the size is 13×13 , the proposed model yields the lowest OA is 98.84%. To be noticed, the larger size of the space contributes to the extraction of spatial features of the data set, and the overall accuracy tends to be stable after 17×17 .

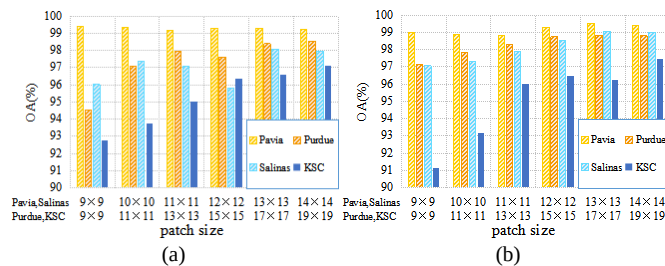


Fig. 13. OA (%) obtained by the proposed network with the different patch size of training samples on three data sets. (a) 2D-3D-D-S (b) 2D-3D-D

F. Effect of Different Numbers of Training Samples

Next, we explore the effect of the different percentages of the training sample with the proposed CNN network in this part. For the Purdue and KSC data, 1% to 10% with 1% interval of the total sample is set in this experiment, for the Salinas image, the portion changes from 1% to 5% with the 0.5% step. The dropout rate of the three sets are all set to 0.3, the sample size of the Purdue and KSC data sets are fixed as 19×19 , and the sample size of the Salinas data is 13×13 . Fig 14 presents the average OA results of the different portions of the training sample which are executed five times separately. It can be seen that the proposed CNN network generates robust performance even the percentage is set to 1%. The performances of the Purdue and KSC generally improve with the sample percentage increases, while for the Salinas Valley data, the OA is more stable because of the characteristic of the rich spatial information. Especially, in the simplified 2D-3D-D model, the lowest OA is 96.65% when the percentage is 3.5% for the Salinas data set, and the best OA can reach 97.98% and 97.14% when the percentage is 10% for the Purdue and the KSC data specific. The 2D-3D-D network generates the best OA of 98.51% with the ratio is 9% for the Purdue data, the more percentage brings the better performance for the rest three data sets with the value of 99.07%, 97.47%, and 99.54%, for the Salinas data, the KSC data and the Pavia data.

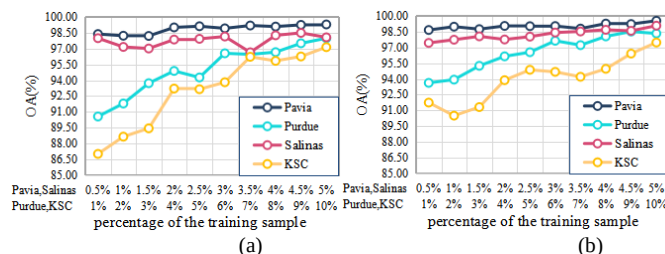


Fig. 14. OA (%) obtained by the proposed network with different percentages of training samples on three data sets. (a) 2D-3D-D-S (b) 2D-3D-D

G. Comparison of the dropout rate

Lastly, the effectiveness of the setting of the dropout rate is analyzed in this section. The proposed 2D-3D-D networks are compared with different dropout percentages. The size and the percentage of the training sample Purdue data and the KSC data is set to 19×19 and 10%, and for Salinas Valley, they are set to 13×13 and 5%. OA obtained by the 2D-3D-D network with different dropout rates on three data sets is demonstrated in Fig. 15, it can be observed that OA varies with the different dropout rate in the 2D-3D-D-S, it leads the best OA for the Purdue data is 97.98% when the rate is 0.3, the best OA of Salinas data is 98.80% when the value is set to 0.8, and the KSC data, the OA of 97.32% when the dropout rate is 0.2. On the contrary, the proposed framework generates the lowest value for Purdue data when the rate is 0.8, for the Salinas data with the rate of 0.6, and for the KSC data when the drop rate is 0.9, it generates the worst OA. The best OA in the 2D-3D-D model for the Purdue data and the Salinas data are 98.33% and 99.07% when the rate is 0.8, and when the dropout rate is 0.3, the model generates the best OA of 97.48%, and the best OA is 99.58% for the Pavia data with the rate of 0.4. It also can be concluded that the dropout rate will not always improve the performance of the training procedure. We also noted that, due to the simplification of the model, the proposed model can achieve relatively great overall accuracy with the bigger dropout rates.

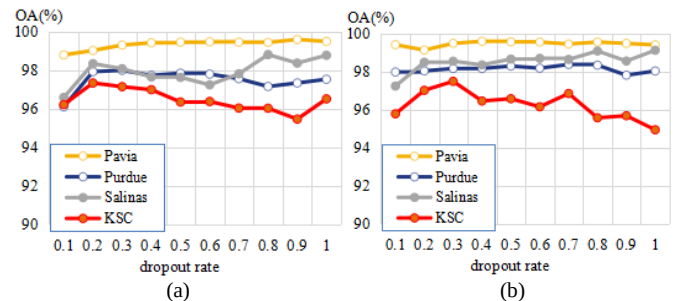


Fig. 15. OA (%) obtained by the 2D-3D-D network with the different dropout rates of training samples on three data sets. (a) 2D-3D-D-S (b) 2D-3D-D

H. Analysis of Number of the Modules

To further analyze the influence of the number of the 2D-3D-D module, we complete a series of experiments to compare the classification performance and the running time of the proposed method with different modules, the programming environment is Nvidia Geforce GTX 1060Ti, RAM 32.0GB for the four data sets. In this section, one 2D block and one 3D block is described as one module. To conduct a fair evaluation, the experiments are implemented on all the four data sets and Table 17-19 listed the classification accuracy of each class. According to the reported results in Table 16- 19, the proposed networks with different modules achieve stable performance for every data set. Table 21 shows the training time of the proposed frameworks (both simplified 3D kernel and 3D kernel) with three different modules separately, and it can be seen that the more modules consume more time cost. Moreover,

as can be seen in Fig. 16 and Fig.17, with the same setting of the hyper-parameters, the proposed architecture with one module obtains the best OA value with the four data sets. It can be concluded that the proposed CNN network with one module is superior to the other modules in terms of classification performance and quantitative time cost.

Table 17 OA calculated from the classification results of Purdue Indiana Indian Pines Scene with different modules of the proposed method with different modules

Class P _{OA} %	2D-3D-D-S			2D-3D-D		
	One Module	Two Modules	Three Modules	One Module	Two Modules	Three Modules
1	99.55	100	1	100	100	100
2	97.61	97.56	96.78	98.36	98.92	95.04
3	96.62	94.99	96.68	97.80	96.75	93.69
4	98.75	98.71	98.33	97.20	97.53	99.16
5	99.67	98.33	97.15	99.30	99.18	96.96
6	99.34	98.64	99.31	99.07	99.45	95.30
7	100	100	96.43	100	100	100
8	99.46	100	99.79	99.83	100	100
9	95.42	95.24	86.96	92.72	94.74	76.98
10	97.35	95.53	96.02	97.34	96.35	97.91
11	97.73	96.10	98.16	98.23	97.75	97.06
12	96.79	94.79	91.97	97.66	96.74	92.80
13	99.42	98.56	100	99.32	100	100
14	98.71	98.44	98.90	99.01	99.68	95.79
15	98.23	95.08	96.73	98.60	98.96	94.18
16	94.92	93.81	97.87	92.59	94.68	91.75
POA	97.98	96.91	97.43	98.33	98.23	96.11
AA	98.10	97.24	90.76	97.94	98.17	95.41

Table 18 OA calculated from the classification results of the Salinas valley with different modules of the proposed method with different modules

Class P _{OA} %	2D-3D-D			2D-3D-D		
	One Module	Two Modules	Three Modules	One Module	Two Modules	Three Modules
1	99.43	99.96	99.00	99.81	96.81	100
2	99.69	99.78	99.52	99.65	99.05	98.65
3	99.50	98.21	98.85	99.75	99.95	100
4	99.06	99.00	95.22	99.37	99.35	96.94
5	99.25	99.45	98.66	98.68	98.40	99.58
6	99.87	99.97	99.86	99.99	100	99.97
7	99.60	99.38	99.78	99.88	99.52	99.25
8	97.17	95.01	97.54	98.05	96.28	96.35
9	99.82	99.49	99.86	99.80	99.37	99.60
10	98.85	99.35	98.60	99.86	99.51	99.75
11	96.61	97.83	97.13	98.67	96.21	93.36
12	98.83	99.80	99.23	99.92	98.77	97.32
13	98.96	99.62	99.60	99.89	100	99.13
14	98.72	99.02	98.71	99.40	99.26	99.91
15	93.52	95.86	87.03	97.76	98.43	90.82
16	99.72	99.05	99.50	99.88	98.62	99.21
POA	98.07	97.99	96.84	99.07	98.40	97.39
AA	98.66	98.89	98.01	99.40	98.72	98.12

Table 19 OA calculated from the classification results of Kennedy Space Center with different modules of the proposed method with different modules

Class P _{OA} %	2D-3D-D-S			2D-3D-D		
	One Module	Two Modules	Three Modules	One Module	Two Modules	Three Modules
1	97.06	89.74	88.14	98.19	96.57	95.84
2	95.52	98.24	93.09	96.75	93.53	97.77
3	97.33	96.88	98.15	98.31	96.36	98.07
4	89.79	91.60	89.50	87.50	84.53	79.73
5	84.59	85.39	79.12	94.10	89.44	85.14
6	99.61	97.24	96.97	98.73	98.10	99.07
7	94.82	86.87	93.06	95.87	100	80.95
8	98.29	97.76	93.19	96.32	90.85	92.44
9	97.44	91.95	91.18	97.34	99.41	99.60
10	97.89	94.87	95.30	97.35	91.69	93.80
11	99.88	97.19	98.53	99.48	99.29	96.77
12	97.19	93.68	96.25	98.90	97.62	95.56
13	99.55	98.10	96.56	99.70	99.25	97.17
POA	97.14	93.84	93.32	97.47	95.85	94.68
AA	96.07	93.81	93.00	96.81	95.13	93.22

Table 20 OA calculated from the classification results of University of Pavia with different modules of the proposed method with different modules

Class P _{OA} %	2D-3D-D-S			2D-3D-D		
	One Module	Two Modules	Three Modules	One Module	Two Modules	Three Modules
1	98.79	98.81	97.91	99.42	98.16	95.20
2	99.91	99.81	99.27	99.93	99.79	98.91
3	97.6	91.29	91.72	98.69	93.18	79.29
4	99.81	99.51	99.50	99.88	100	97.33
5	99.75	98.66	99.19	99.97	99.70	99.85
6	99.18	98.91	98.72	99.45	99.66	98.99
7	99.47	94.90	94.56	99.47	98.41	86.72
8	97.48	95.12	94.72	97.89	92.93	92.45
9	99.68	98.77	99.32	99.96	99.89	99.05
POA	99.29	98.45	98.07	99.54	98.56	96.36
AA	99.074	97.31	97.21	99.41	97.97	94.20

Table 21 The training time of the proposed CNN frameworks with different modules of the four data sets (Seconds)

Time (Seconds)	2D-3D-D-S			2D-3D-D		
	One module	Two modules	Three modules	One module	Two modules	Three modules
Purdue	270	305	352	530	629	681
Salinas	352	471	589	557	704	892
KSC	474	564	668	980	1257	1440
Pavia	192	375	358	354	518	669

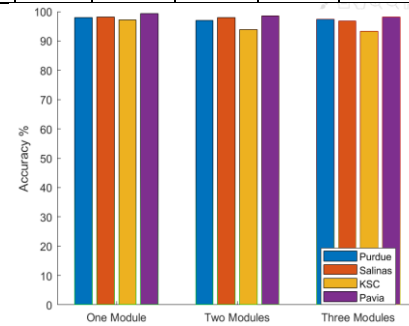


Fig. 16. OA (%) obtained by the proposed 2D-3D-D-S networks with different modules of the four data sets

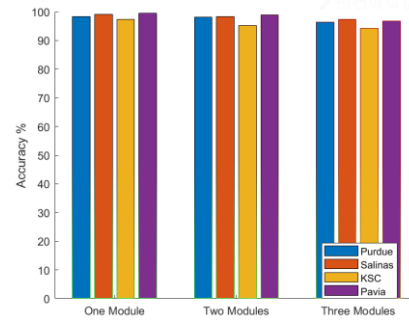


Fig. 17. OA (%) obtained by the proposed 2D-3D-D networks with different modules of the four data sets

V. CONCLUSIONS

In this study, we propose a new network model for hyperspectral image classification based on the cooperation between 2D CNN and simplified 3D convolution layer. The 2D CNN part focuses on the extraction of rich spectral-spatial features from the available HSI. Subsequently, the 3D block mainly deals with the reconstruction of the refined spectral feature with neighbor bands information involved. In this way, the proposed model can generate the fusion refined feature to enhance the representation and description of the deep map. The key role of this framework is that the 3D block depends on only one convolution kernel with a size of $1 \times 1 \times L$ to increase the convolution speed and overcome the fitting problem. The proposed model explored an implementation with the depth-wise separable convolution way for HSIC for the first

time. The real hyperspectral image experimental results proved that the proposed architecture achieves a great performance on the four popular testing data sets. Notice that the band information is crucial for feature refinement in the CNN network for HSIC, one of our future work will focus on design band selection and augmentation network to extract contextual feature hierarchically.

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